

Chain Matrix Multiplication :

1. Definition:

Chain Matrix Multiplication is a dynamic programming technique used to find the best order for multiplying a sequence of matrices so that the number of scalar multiplications is minimum.

The matrices are not changed; only their **multiplication order** is changed.

Algorithm:

Step 1: Read the number of matrices n .

Step 2: Read the dimensions of all matrices and store them in array p .

Step 3: Set the cost of multiplying one matrix to 0.

$$m[i][i] = 0$$

Step 4: Consider chains of length 2, then 3, and so on up to n .

Step 5: For every chain, try every possible position k to split the chain.

Step 6: Calculate the cost using:

$$\text{cost} = m[i][k] + m[k+1][j] + p[i-1] \times p[k] \times p[j]$$

Step 7: Select the minimum cost among all possible splits.

Step 8: Finally, $m[1][n]$ gives the minimum number of scalar multiplications.

Time Complexity:

Case	Time Complexity
Best Case	$O(n^3)$
Average Case	$O(n^3)$
Worst Case	$O(n^3)$

Space Complexity:

Space Complexity = $O(n^2)$

OUTPUT:

Output

```
Enter number of matrices: 3
Enter the dimensions:
10
20
30
40
Minimum number of multiplications = 18000
```

```
=== Code Execution Successful ===
```

Conclusion:

The Chain Matrix Multiplication problem was successfully implemented using Dynamic Programming. The algorithm finds the optimal order of multiplying matrices by checking different possible arrangements and selecting the one with minimum scalar multiplication cost. Thus, Dynamic Programming helps to reduce repeated calculations and efficiently find the minimum multiplication cost.